

ECE113: Basic Electronics (BE)
Winter 2025

Mid Semester Exam

Date: 27-Feb-2025

Duration: 2 Hours

Total Marks: 40

[CO1, CO2] Q1: [10 Marks] Find the value of V_x and V_y by using Superposition theorem (in Figure-1).

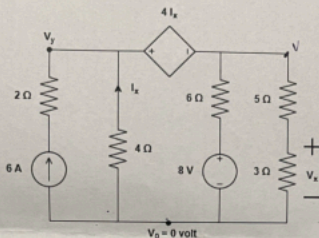


Figure 1

$$\begin{aligned} \frac{8-V}{63} &= \frac{V-0}{84} \\ 32-4V &= 3V \\ 32 &= 7V \\ V &= \frac{32}{7} \\ S &= \frac{32}{7} = \frac{4}{7} \end{aligned}$$

[CO3] Q2: [10 Marks] In the circuit shown (in Figure-2) at time $t = 0$ sec the switch (S) is connected to position-A. After one time constant ($T=2$ sec) the switch (S) is transferred to position-B.

- (a) Voltage response across capacitor, when the switch is connected to position-A
(b) Current response through capacitor, when the switch is connected to position-B.

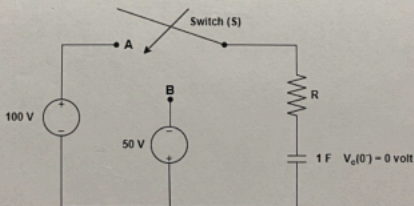


Figure 2

[CO3] Q3: [10 Marks] In Figure-3, find-

- (a) $i_1(0^+)$ (b) $i_2(0^+)$ (c) $V_c(0^+)$ (d) $\frac{di_1(0^+)}{dt}$ (e) $\frac{di_2(0^+)}{dt}$
(f) $\frac{d^2i_1(0^+)}{dt^2}$ (g) $\frac{d^2i_2(0^+)}{dt^2}$

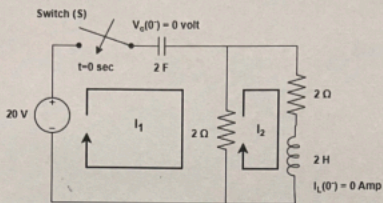


Figure 3

[CO1, CO2] Q4: [5 Marks] Find the value of current (I_0) (in Figure-4).

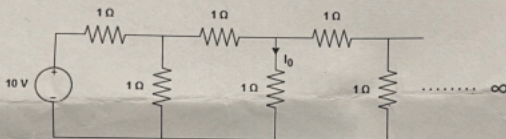


Figure 4

[CO1, CO2] Q5: [5 Marks] Find the value of resistance "R" to obtain maximum power transfer from Network-A to Network-B (in Figure-5) at steady state.

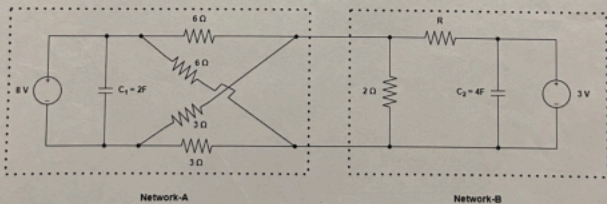


Figure 5